

RACHNA KADAM

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SUMMARY

Robotics-focused NYU M.S. candidate with hands-on experience building real-world autonomous systems across perception, planning/control, and embedded robotics (ROS 2, computer vision, and microcontrollers).

EXPERIENCE

- Research Assistant - NYU Mechatronics, Controls and Robotics Lab / Dr. Vikram Kapila** **June 2025 - Present**
VLM-LLM Integrated Multi-Robot System for Autonomous Task Execution
- Built a zero-code mobile robotics system that converts natural-language tasks into real robot execution through an LLM/VLM-driven perception → planning → control pipeline.
 - Implemented overhead-camera localization with ArUco/Charuco calibration and multi-camera homography stitching (validated on 3 cameras, scalable), plus dynamic obstacle detection and RRT* navigation with online replanning.
 - Designed a modular system architecture using RPC (ZeroMQ + CBOR) for robot/task services and integrated Pure Pursuit + PID docking control with a custom ESP32 wireless command bridge.
- Course Assistant – Automatic Controls Laboratory** **Sep 2025 – Present**
New York University, Tandon School of Engineering
- Supported Automatic Controls labs with MATLAB/Simulink + Quanser QuaRC, implementing/debugging real-time PID/LQR on DC motor, maglev, rotary pendulum, and coupled-tank plants.
 - Developed new Arduino-based labs on embedded I/O (ADC/DAC/PWM), digital filtering, and model-based thermal control.
 - Built a thermal control experiment using step tests, FOPDT system ID, and discrete control validation.
 - Implemented real-time filtering for IMU noise suppression using discretized Butterworth low-pass filters.
- AI Systems Engineer** **Dec 2023 – May 2024**
Lawyantra, Mumbai, India
- Engineered a high-performance NLP pipeline using RAG architecture and TF-BERT, optimizing inference time for large-scale data retrieval.
 - Built a bespoke legal search engine with semantic and keyword capabilities, incorporating advanced retrieval algorithms such as TF-BERT, BM25, TF-IDF to improve document ranking.
 - Optimized Generative AI models like LLaMA and Mistral for high-precision summarization by fine-tuning parameters, enhancing summary accuracy, and reducing storage overhead using Hugging Face frameworks.
- Mechanical Design & Validation Engineer (SAE BAJA / Student Team)** **Sep 2021 – May 2022**
MSPTME Racing Team
- Designed, validated, and tested the hydraulic braking circuit for an All-Terrain Vehicle (ATV) using CAD software, including Fusion360 and SolidWorks, achieving precise functionality and robust design.
 - Executed comprehensive validation processes utilizing Ansys to confirm simulation results, meeting rigorous performance criteria across 15 testing conditions while enhancing reliability in high stress driving situations during field trials.
 - Assembled and tested the braking subsystem as a vehicle-level system integration effort, validating hydraulic and mechanical-mechatronic components under competitive field trials and safety constraints.
 - Achieved 10th rank in Phase 2 (Sales) of SAE BAJA, demonstrating exceptional presentation and business strategy skills in promoting the ATV design to industry experts and judges.

PROJECTS

- Interactive Motion-Controlled Rehabilitation Game Using Computer Vision and IMUs** **April 2025– May 2025**
- Designed and rapidly prototyped a real-time interactive mechatronic system using OpenCV + IMUs to detect punch direction/intensity via sensor fusion.
 - Integrated Raspberry Pi, Arduino Nano, and ESP32 for wireless data flow between vision and inertial modules, achieving < 10ms latency.
 - Developed the game engine and UI using PyGame, enabling quadrant-based scoring and dynamic feedback.
 - Overcame hardware bottlenecks through optimized processing; laid groundwork for multiplayer and fitness-tracking extensions.
- Indoor Mapping Robot with Custom SONAR and Real-Time ROS Visualization** **Jan 2025 - Feb 2025**
- Developed a differential drive robot for indoor environment visualization, achieving a localization accuracy of ± 2 mm using an IMU and magnetic encoders.
 - Designed a custom system with an ultrasonic sensor, generating a point cloud with a resolution of 1 mm per point for mapping applications.
 - Implemented real-time data transmission and visualization via ROS, reducing data processing latency by 30% and enabling efficient autonomous navigation and indoor surveying.

EDUCATION

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| New York University, New York, NY
<i>Master of Science, Mechatronics and Robotics</i> | May 2026
GPA: 3.61 |
| NMIMS University, Mumbai, India
<i>Bachelor of Technology, Mechatronics</i> | May 2024
GPA: 3.69 |

SKILLS

- Languages:** Python, C/C++, MATLAB
- Robotics:** ROS 2, Localization, SLAM, Path Planning, Kinematics & Dynamics, Sensor Fusion (IMU/Encoders/Vision)
- AI & Perception:** OpenCV, ArUco Detection, PyTorch, Vision-Language Models, LLM Integration, RAG, Transformers (BERT)
- Controls:** PID, LQR, System Identification, State Estimation, Model-Based Design, Real-Time Feedback Control
- Embedded Systems:** ESP32, Arduino, Raspberry Pi, Jetson; I2C/UART, SPI, Wi-Fi, BLE, ESP-NOW; PLC (Ladder Logic—IndraWorks/Bosch Rexroth), SCADA/HMI
- Tools & CAD/CAE:** Git/GitHub, Linux (Ubuntu), SolidWorks, Fusion 360, Ansys, 3D Printing, Laser Cutting